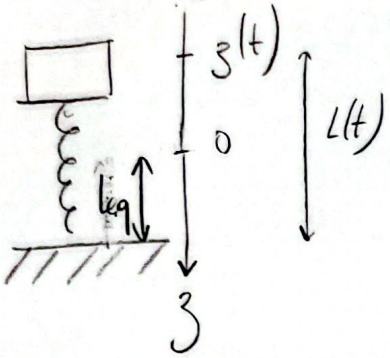
Interro 5 - sujet B



1) BDF $\vec{P} = +mg\vec{u_z}$

$\vec{f_r} = -k(L(t) - L_0)\vec{u_z}$

$= +k(L(t) - L_0)\vec{u_z}$

avec $\boxed{L(t) = L_{eq} - z(t)}$

$\vec{f_r} = +k(L_{eq} - z(t) - L_0)\vec{u_z}$

$\vec{f} = -\lambda \dot{z}\vec{u_z}$

2) En statique : $mg + k(L_{eq} - L_0) = 0 \iff \boxed{-\frac{mg}{k} + L_0 = L_{eq}}$

3) PFD : $\sum \vec{F} = m\vec{a}$

$mg + k(L_{eq} - z(t) - L_0) - \lambda \dot{z} = m\ddot{z}$

$\iff mg + k\left(-\frac{mg}{k} + L_0 - z(t) - L_0\right) - \lambda \dot{z} = m\ddot{z}$

$\iff \ddot{z} + \frac{\lambda}{m}\dot{z} + \frac{k}{m}z = 0$

4) $\omega_0 = \sqrt{\frac{k}{m}}$ $\quad \frac{\omega_0}{Q} = \frac{\lambda}{m} \iff Q = \omega_0 \times \frac{m}{\lambda} = \boxed{2_0} > \frac{1}{2}$

4) solution de l'ED : $z(t) = e^{-t/2}(A\cos(\Omega t) + B\sin(\Omega t))$

conditions initiales

$z(t=0) = 0 \iff \boxed{A = 0}$

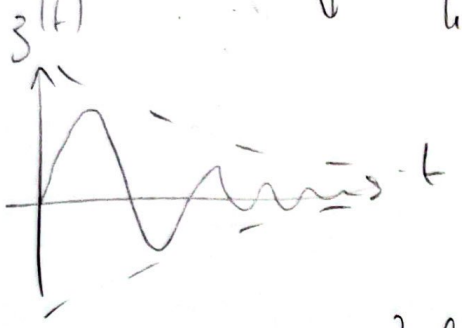
$\dot{z}(t=0) = z_0 \qquad \dot{z}(t) = -\frac{1}{2}e^{-t/2}B\sin(\Omega t) + e^{-t/}$

$\iff \dot{z}(t=0) = B\Omega = v_0 \iff \boxed{B = \frac{v_0}{\Omega}}$

$\boxed{z(t) = \frac{v_0}{\Omega}\sin(\Omega t)e^{-t/2}}$

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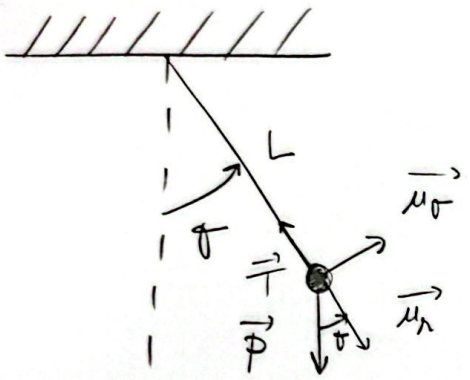
5) $T = \frac{2\pi}{\Omega} = \frac{2\pi}{\omega_0 \sqrt{1 - \frac{1}{4Q^2}}} = \boxed{0{,}314 \text{ s}}$

6) 

7) $N = \frac{t}{T} = \frac{-2 \ln\left(5 \times 10^{-3} \times \frac{\Omega}{\omega_0}\right)}{T} = \boxed{5{,}83 \text{ pseudo-T}}$

Interro A - sujet A

1)



Système = masse

BDF : $\vec{P} = mg\cos(\theta)\vec{u_r} - mg\sin(\theta)\vec{u_\theta}$

$\vec{T} = -T\vec{u_r}$

$\vec{f} = -\lambda\vec{v}$

En coordonnées polaires, on a : $\vec{OM} = L\vec{u_r}$

$\vec{v} = L\dot{\theta}\vec{u_\theta}$

$\vec{a} = L\ddot{\theta}\vec{u_\theta} - L\dot{\theta}^2\vec{u_r}$

soit $\vec{f} = -\lambda L\dot{\theta}\vec{u_\theta}$

2) PFD : $\sum\vec{F} = m\vec{a}$

$\vec{P} + \vec{f} + \vec{T} = m(L\ddot{\theta}\vec{u_\theta} - L\dot{\theta}^2\vec{u_r})$

selon $\vec{u_\theta}$: $-mg\sin(\theta) - \lambda L\dot{\theta} = mL\ddot{\theta}$

Dans l'approximation des petits angles $\sin(\theta) \simeq \theta$

soit : $\boxed{\ddot{\theta} + \frac{\lambda}{m}\dot{\theta} + \frac{g}{L}\theta = 0}$

3) $\omega_0 = \sqrt{\frac{g}{L}}$; $\frac{\omega_0}{Q} = \frac{\lambda}{m} \iff Q = \frac{m}{\lambda} \times \omega_0 = \frac{m}{\lambda} \times \sqrt{\frac{g}{L}}$

$= \boxed{3{,}61 \text{ rad/s}}$

$= \frac{0{,}2}{5\times10^{-2}} \times \sqrt{\frac{9{,}81}{0{,}75}}$

$\boxed{Q = 14{,}46}$

4) Pour $Q > 1/2$, la solution homogène est de la forme :

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$$\theta(t) = e^{-t/2}\left(A\cos(\Omega t) + B\sin(\Omega t)\right)$$

Conditions initiales :

$$\theta(t=0) = 0 \iff \boxed{A = 0}$$

$$\dot{\theta}(t=0) = \omega_i$$

$$\dot{\theta}(t) = -\frac{1}{2}e^{-t/2}\,B\sin(\Omega t) + e^{-t/2}\left(B\Omega\cos(\Omega t)\right)$$

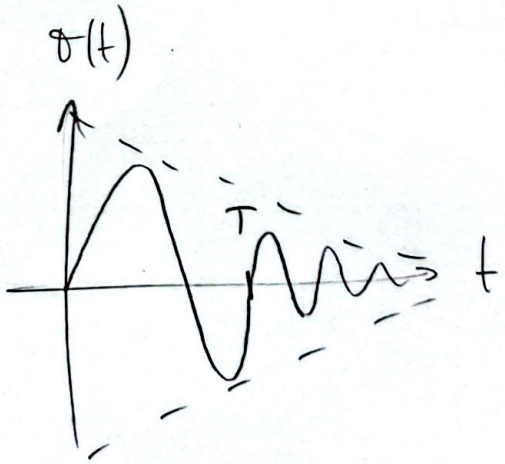
$$= Be^{-t/2}\left(-\frac{1}{2}\sin(\Omega t) + \Omega\cos(\Omega t)\right)$$

$$\dot{\theta}(t=0) = B\Omega = \omega_i \iff \boxed{B = \frac{\omega_i}{\Omega}}$$

soit $\boxed{\theta(t) = \frac{\omega_i}{\Omega}e^{-t/2}\sin(\Omega t)}$

5) $$T = \frac{2\pi}{\Omega} = \frac{2\pi}{\omega_0\sqrt{1-\frac{1}{4Q^2}}} = \frac{2\pi}{\sqrt{1-\frac{1}{4\times 14{,}46^2}}} = \boxed{1{,}73\,s}$$

6)



7) $$\frac{\omega_i}{\Omega}e^{-t/2} = 1^\circ$$

$$N = \frac{t}{T} = \frac{-2\ln\left(1^\circ\times\frac{\Omega}{\omega_i}\right)}{}$$

$$= \boxed{10{,}35 \text{ pseudo}}$$